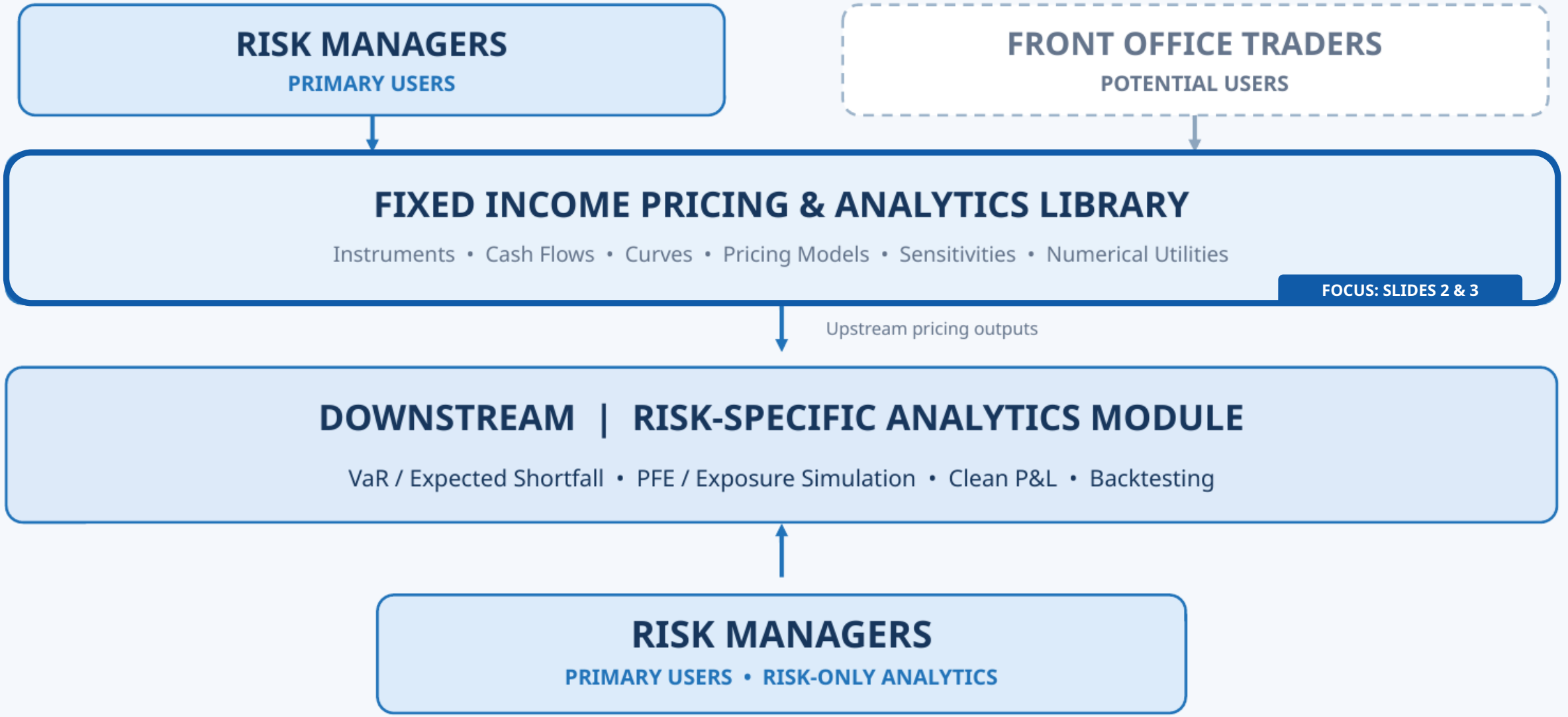
Fixed Income Analytics Library - Proposed Architecture

Architecture Overview



Fixed Income Analytics Library - Proposed Architecture

Code Structure (1/2)

fixed_income_library/ | Instrument, Cash Flow Generation & Market Data

▱ instrument/

- ▤ fixed_income_instrument.py
- ▱ **bond/**
  - ▤ bond.py
  - ▤ fixed_rate.py
  - ▤ floating_rate.py
  - ▤ inflation.py
  - ▤ fixed_to_float.py
  - ▱ **api/**
    - ▤ bond_builder.py
    - ▤ yield_discount_margin.py
    - ▤ spreads_calculation.py
    - ▤ yield_based_risk.py
    - ▤ spread_based_risk.py
- ▱ **swap/**
- ▱ **futures/**
- ▱ **rate_lock/**
- ▱ **cap_floor/**
- ▱ **swaption/**

Includes linear and option derivatives

▱ cash_flow_generation/

- ▤ cash_flow_generator.py
- ▤ coupon_schedule.py
- ▤ principal_schedule.py
- ▤ cash_flow_display.py

Independent top-level module

Reusable across bonds, swaps and derivatives

▱ market_data/

- ▱ **curve/**
  - ▱ **construction/**
    - ▤ bootstrap.py
    - ▤ optimization.py
  - ▱ **bumping/**
    - ▤ parallel.py
    - ▤ scenario.py
  - ▱ **serialization/**
    - ▤ serialize.py
    - ▤ deserialize.py
- ▱ **vol/**
  - ▱ **model/**
    - ▤ surface.py
    - ▤ sabr.py
  - ▱ **bumping/**
    - ▤ parallel.py
    - ▤ scenario.py
  - ▱ **serialization/**
    - ▤ serialize.py
    - ▤ deserialize.py

Fixed Income Analytics Library - Proposed Architecture

Code Structure (2/2)

fixed_income_library/ | Pricing Models, Numerical & Testing

pricing_models/

- **short_rate/**
  - hull_white_1f.py # Hull-White 1F
  - hull_white_2f.py # Hull-White 2F
  - g2pp.py # Two-factor Gaussian
  - black_karasinski.py # Lognormal short rate
- **market_models/**
  - lmm.py # LIBOR Market Model
  - qg.py # Quadratic Gaussian
- **pricing_engines/**
  - tree.py # Lattice pricing
  - monte_carlo.py # Simulation pricing
  - pde.py # PDE pricing
  - analytic.py # Closed-form pricing

numerical/ (Math)

- **optimization/**
  - levenberg_marquardt.py # Nonlinear calibration
  - least_squares.py # Least-squares fit
- **interpolation/**
  - linear.py # Linear interpolation
  - cubic_spline.py # Cubic spline
- **integration/**
  - quadrature.py # Numerical integration
- **jacobian_transformation/**
  - jacobian.py # Jacobian matrix
  - risk_transformation.py # Risk basis transformation

testing/

- **unit/**
  - test_cashflows.py # Schedules / amounts
  - test_instruments.py # Bond / swap conventions
  - test_numerical.py # Solver / interpolation
- **regression/** # Critical: golden results
  - test_bond_pricing.py # PV / yield / spread
  - test_curve_bootstrap.py # Curve nodes / DF
  - test_risk.py # DV01 / Greeks
  - test_models.py # Callable / stochastic
- **benchmark/**
  - test_quantlib.py # Reference comparison
- **integration/**
  - test_end_to_end.py # Build -> price -> risk
- **fixtures/**
  - market_snapshots.py # Frozen market data
  - golden_results.json # Approved baselines